Code: DM-205 **Paper ID: 12005**

GNIOT Institute of Management Studies – PGDM (Batch 2024-26)

(TRIMESTER –II) END TERM EXAMINATION, JANUARY -2025

Basic Python for Managers

[Time: 2 Hours] ***[Maximum Marks: 60]***

Following are the course outcomes of the subject: - Basic Python for Managers

CO1	Python, including installation, basic syntax, variables, data types, operators, and type conversion, enabling them to write basic Python programs.
CO2	Proficient in designing and implementing control flow structures and functions in Python to solve complex problems.
CO3	Select and apply appropriate data structures in Python, optimizing data organization and manipulation for various computational tasks.
CO4	Manage files and handling exceptions in Python, ensuring efficient data storage and robust error handling in their programs.

(Note- Faculty can add the more CO's as per their subject.)

SECTION-A

Attempt all parts of the following: (Short Answer Type Questions, do not exceed 150- 200 words)

Q.1	Marks	COURSE OUTCOME	BLOOMS LEVEL
(A)	**4**	CO - 1	BL1
(B)	**4**	CO - 1	BL1
(C)	**4**	CO - 4	BL2
(D)	**4**	CO - 1	BL1
(E)	**4**	CO - 1	BL1

SECTION-B

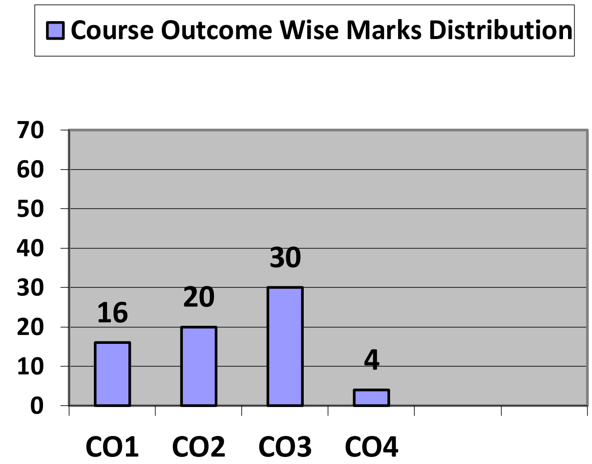
Attempt ANY TWO questions out of the following: (Long Answer Type Questions, do not exceed 300- 400 words)

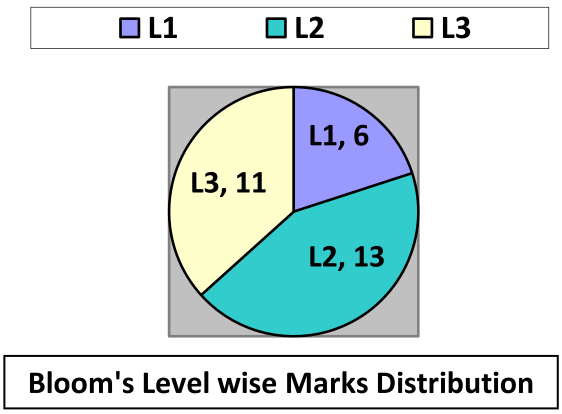
Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
2 (A)	**10**	CO – 2	BL2
2 (B)	**10**	CO – 2	BL2
3 (A)	**10**	CO – 3	BL2
3 (B)	**10**	CO – 3	BL2

SECTION-C

Attempt all parts of the following: (Caselet / Numerical, do not exceed 300- 400 words)

Q.4	Marks	COURSE OUTCOME	BLOOMS LEVEL
4 (A)	**10**	CO – 3	BL2
4 (B)	**10**	CO - 3	BL2





Moderated By **IQAC** **Dean - Examinations**

Printed Pages: 1

Roll No: ☐☐☐☐☐☐☐☐☐☐ ***Subject Code:*** **DM-205**

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(TRIMESTER –II) END TERM EXAMINATION, JANUARY - 2025

Basic Python for Managers

[Time: 2 Hours] ***[Maximum Marks: 60]***

INSTRUCTIONS:

(i) ***There are THREE sections in this question paper.***
(ii) ***Attempt questions from each section as per the directions.***
(iii) ***Make suitable assumptions wherever necessary.***

SECTION-A
(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following: ***(5×4=20)***

a) Define Python. Why is it considered a popular programming language?
b) Define exception handling with example
c) Explain 5 Features of Python.
d) Discuss variables, data types in Python with suitable examples.
e) What are Python operators? Explain the different types of operators in Python with examples.

SECTION-B
(Long Answer Type Questions)

Q.2. Attempt all questions of the following ***(2X10=20)***

(a) Create a tuple with the following elements: `('apple', 'banana', 'cherry', 'date', 'apple', 'fig', 'grape')`.

i. Slice the tuple to display the first 4 elements.
ii. Ask the user to input a fruit name and check if it exists in the tuple. Print an appropriate message.

OR

(b) Create a dictionary with the following key-value pairs:
`{'name': 'John', 'age': 25, 'city': 'New York', 'profession': 'Engineer'}`

i. Add a new key `'salary'` with the value `50000`.
ii. Update the `'age'` key to `26`.

Q.3. *(a)* Write a program that calculates the sum of all odd numbers between 1 and 100. Use a `for` loop to iterate through the numbers.

OR

(b) Write a program that takes a student's score as input and assigns a grade based on the following conditions:

- Score ≥ 90: "A"
- Score ≥ 75 and < 90: "B"
- Score ≥ 50 and < 75: "C"
- Score < 50: "Fail"

SECTION-C

Q.4. Attempt all parts of the following: ***(2×10=20)***

Case Study: Employee Leave Management System

Scenario:

You are developing a leave management system for a company. The system should track and manage employee leaves based on their type and eligibility. Employees are allowed different types of leaves, including **Sick Leave**, **Casual Leave**, and **Annual Leave**. The system should calculate if an employee is eligible for leave and whether the leave can be granted based on the number of leaves available and the employee's remaining balance.

4(a) Write a program that:

1. Takes input for the **employee's name**, **leave type**, **leave days requested**, and whether a **medical certificate** is provided for Sick Leave.
2. Checks if the employee has sufficient leave balance for the requested type of leave.

4(b)

3. Determines if the leave request can be approved based on the available leaves, eligibility, and company rules.
4. Displays the approval status, and if approved, updates the leave balance accordingly.